

Examination paper for TFE4141 Design of Digital System 1

Academic contact during examination: Per Gunnar Kjeldsberg

Phone: 934 59 550

Academic contact present at the exam location: YES (around 11:00 o'clock)

Examination date: Monday December 16

Examination time (from-to): 09:00 – 13:00

Permitted examination support material C: Specified printed and hand-written support material is allowed. A specific basic calculator is allowed.

You are allowed to bring the course textbook (Dally, Harting and Aamodt, "Digital design using VHDL, a systems approach") as well as four hand-written one-sided sheets of paper with your own notes.

Other information:

Read through the question set before you start answering. Plan how much time you should spend on each. If a question is unclear/vague, make your own assumptions and specify them in your answer. The total number of achievable points is 100. The maximum numbers for individual questions are indicated for each.

Language: English

Number of pages (front page excluded): 10

Number of pages enclosed: 1

Informasjon om trykking av eksamensoppgave

Originalen er:

1-sidig ☐ 2-sidig ☐

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skal ha flervalgskjema ☐

Checked by:

Date

Signature

Task 1 thorough 12 are Multiple Choice. Write your answer in the table in Appendix A. Detach the sheet with Appendix A and hand it in as part of your submission.

Task 1 through 6 gives a maximum of 2 point. The remaining gives a maximum of 3 points. For these last six you can include a short explanation of your answer in the regular part of your submission (not in Appendix A). If it is reasonable, you can get up to +1 point instead of -1 even if your answer in Appendix A is wrong.

Task 1: Simulation result

(Correct answer 2 points, wrong answer -1 point, no answer 0 points)

Given the code in Figure 1, which timing diagram shows the correct behavior, A, B, or C?

```
library ieee;
use ieee.std_logic_1164.all;

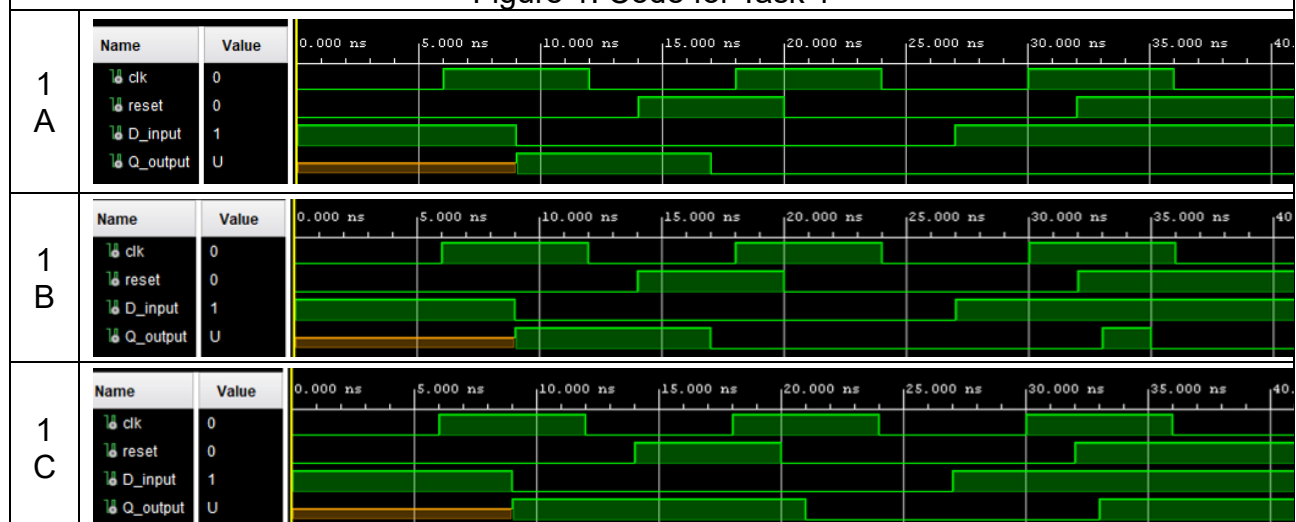
entity Task1_24 is
    port (
        clk      : in std_logic;
        reset     : in std_logic;
        D_input   : in std_logic;
        Q_output  : out std_logic
    );
end Task1_24;

architecture rtl of Task1_24 is
begin

    process(clk, reset) begin
        if(reset='1') then
            Q_output <= '0' after 3ns;
        elsif(clk'event and clk='1') then
            Q_output <= D_input after 3ns;
        end if;
    end process;

end rtl;
```

Figure 1: Code for Task 1



Task 2: Matching block diagram and VHDL code

(Correct answer 2 points, wrong answer -1 point, no answer 0 points)

Given the code and symbol definition in Figure 2, which block diagram matches the code, A, B, or C?

```

library ieee;
use ieee.std_logic_1164.all;
use ieee.numeric_std.all;

entity Task2_24 is
  port( clk : in std_logic;
        a, b, c, d: in unsigned(7 downto 0);
        y : out unsigned(7 downto 0) );
end Task2_24;

architecture rtl of Task2_24 is
  signal sum_ab2: unsigned(7 downto 0);
  signal sum_cd2: unsigned(7 downto 0);
begin

  process(clk) is
    variable sum_ab1: unsigned(7 downto 0);
    variable sum_cd1: unsigned(7 downto 0);
  begin
    if(clk'event and clk = '1')
    then
      sum_ab1 := a + b;
      sum_ab2 <= sum_ab1;
      sum_cd2 <= sum_cd1;
      sum_cd1 := c + d;
    end if;
  end process;

  y <= sum_ab2 + sum_cd2;

end rtl;

```

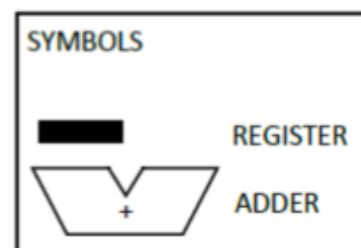
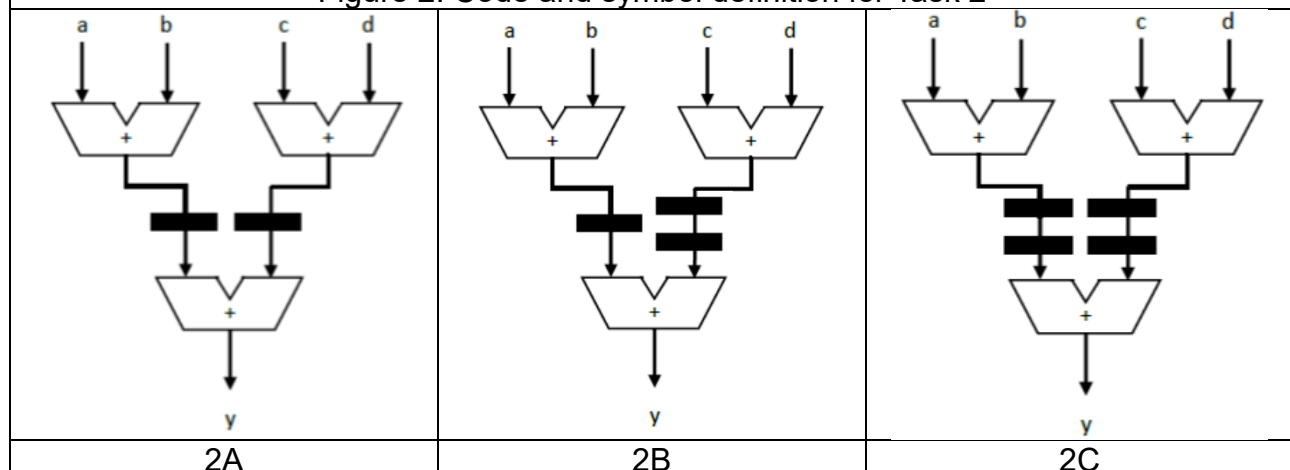


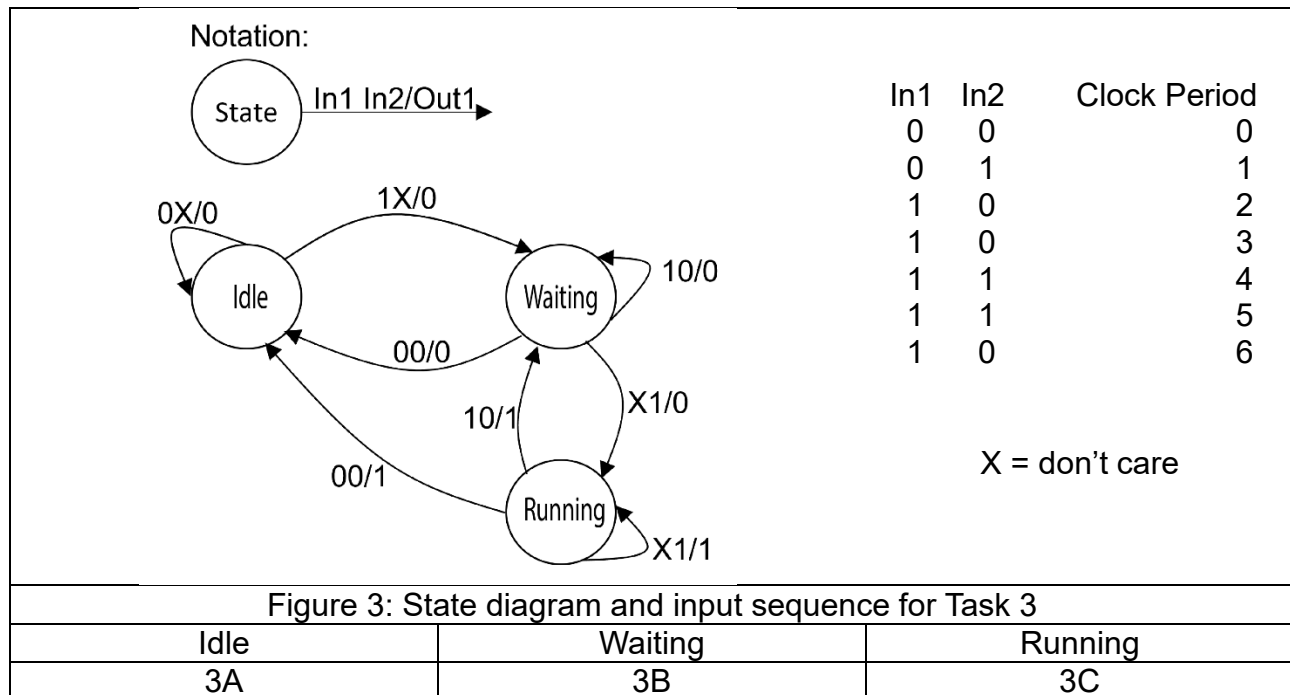
Figure 2: Code and symbol definition for Task 2



Task 3: State Machine

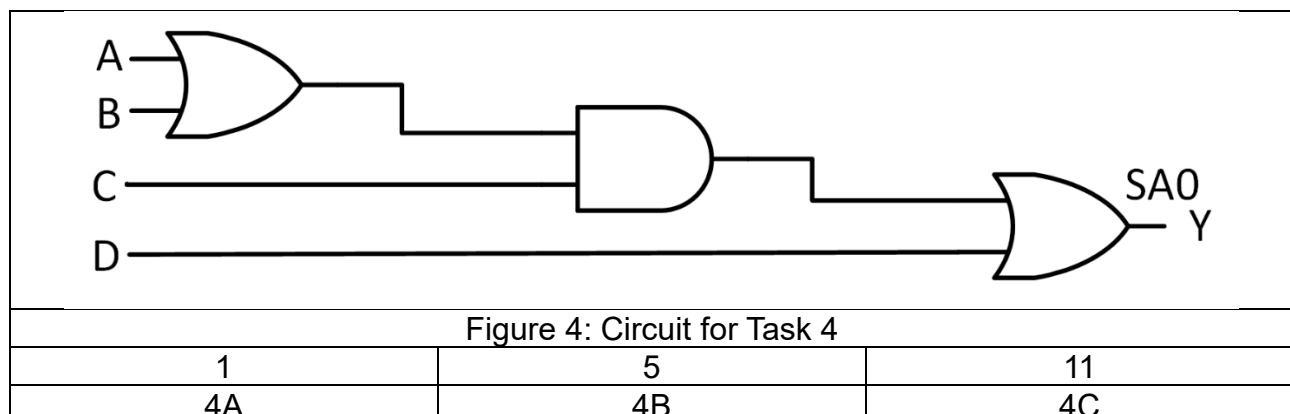
(Correct answer 2 points, wrong answer -1 point, no answer 0 points)

Given the state diagram and the sequence of input signals in Figure 3. Assume that In1 and In2 change value on the negative flank of a clock, while the state machine changes state on the positive flank of the same clock. All setup and hold times are complied with. Assume further that the state machine starts in state Idle. In what state do we end up after the final input values in Clock Period 6 has been evaluated by the state machine, A Idle, B Waiting, or C Running?

**Task 4: Test pattern**

(Correct answer 2 points, wrong answer -1 point, no answer 0 points)

Given the circuit in Figure 4. How many different test patterns on the primary inputs A through E can detect the primary output Y SA0, A 1, B 5, or C 11?



Task 5: Result from synthesis

(Correct answer 2 points, wrong answer -1 point, no answer 0 points)

Given the code in Figure 5. What type of circuit will result from synthesis of this process, A Combinatorial, B Sequential with at least one flip-flop, or C Sequential with at least one latch?

```
process(a,b,c,d,e) begin
  if(a='1') then
    y <= '1';
  elsif(b='0') then
    y <= '0';
  elsif(c=d) then
    y <= '1';
  elsif(e='0') then
    y <= '0';
  end if;
end process;
```

Figure 5: Code for Task 5

Combinatorial	Sequential with at least one flip-flop	Sequential with at least one latch
5A	5B	5C

Task 6: Resulting signal value

(Correct answer 2 points, wrong answer -1 point, no answer 0 points)

Assume that signal Y in Figure 6 is declared with the correct type. If this is a legal statement, what is the resulting value on Y, A the code line contains a syntax error, B '0', or C "1101"?

```
Y <= '1' & '1' & '0' & '1';
```

Figure 6: Code line for Task 7

Syntax error	'0'	"1101"
6A	6B	6C

Task 7: Timing calculation

(Correct answer 3 points, wrong answer -1 point, no answer 0 points)

You are going to implement the polynomial function $Y = X^3 - X^2 + X - 1$. Due to chip area constraints, you can use maximum two multiplier and two combined adder/subtractor modules. The multiplier has a critical path of 60ns while the combined adder/subtractor has a critical path of 40ns. If it is possible to implement, what is the minimum delay

through a fully combinatorial circuit realizing the polynomial function, A Not possible, B 100ns, C 160ns, or D 200ns?

Polynomial function: $Y = X^3 - X^2 + X - 1$		Available resources: * * +/- +/-	
Figure 7: Function and resources for Task 7			
Not possible	100ns	160ns	200ns
7A	7B	7C	7D

Task 8: Energy consumption

(Correct answer 3 points, wrong answer -1 point, no answer 0 points)

You need to estimate how much energy a chip consumes each time a multiplication is performed. You have the information in Figure 8 about the chip and multiplier module. Use the information you need to calculate if the estimated energy consumption is A 2.6pJ, B 14.4pJ, C 55.4pJ, or D 110.2pJ?

Supply voltage 1.2V Clock frequency: 150MHz Number of transistors in multiplier: 1 000 Average fan-out from each transistor output to input gates of other transistors: 2 Average transition activity on each transistor output during one multiplication (0->1 or 1->0): 0.8 Average transistor input gate capacitance: 10fF Average interconnect and parasitic capacitances of each transistor output: 5fF Note: $f = 10^{-15}$, $p = 10^{-12}$			
Figure 8: Information about chip and multiplier in Task 8			
2.6pJ	14.4pJ	55.4pJ	110.2pJ
8A	8B	8C	8D

Task 9: True/false statements

(Correct answer 3 points, wrong answer -1 point, no answer 0 points)

How many of the following three statements are true, A 0, B 1, C 2, or D 3?

- PPA requirements capture the functional aspects of a design.
- The order of VHDL code lines written directly inside an architecture influences the simulation or synthesis result.
- If two sources drive the same std_logic signal to '1' and 'W', respectively, the resulting signal value will be 'W'.

0 are true	1 is true	2 are true	3 are true
9A	9B	9C	9D

Task 10: True/false statements

(Correct answer 3 points, wrong answer -1 point, no answer 0 points)

How many of the following three statements are true, A 0, B 1, C 2, or D 3?

- i. VHDL allows overloading, which means that we can have several different implementations of, e.g., the + operator.
- ii. The statement $A \leq B \leq C$; contains two relational operators.
- iii. You should not have more than one wait statement in a process if it is going to be synthesizable.

0 are true	1 is true	2 are true	3 are true
10A	10B	10C	10D

Task 11: True/false statements

(Correct answer 3 points, wrong answer -1 point, no answer 0 points)

How many of the following three statements are true, A 0, B 1, C 2, or D 3?

- i. Simulation in VHDL using signals is normally faster than simulation using variables.
- ii. If you instantiate components in your VHDL code, you will in addition use a separate component configuration.
- iii. In VHDL, all procedures require a return statement.

0 are true	1 is true	2 are true	3 are true
11A	11B	11C	11D

Task 12: True/false statements

(Correct answer 3 points, wrong answer -1 point, no answer 0 points)

How many of the following three statements are true, A 0, B 1, C 2, or D 3?

- i. In VHDL, assertions can be used to report error conditions during simulation.
- ii. In advanced testbench design, verbosity control can be used to limit the size of the input data set used during simulation.
- iii. In standard cell libraries you can find cells consisting of multiple gates in order to reduce cell area and interconnect requirements.

0 are true	1 is true	2 are true	3 are true
12A	12B	12C	12D

REMEMBER that for task 7 through 12 you can include a short explanation of your answer in addition to using the table in the Appendix. See Page 1 for details.

Task 13: Reduced energy consumption (11 points)

Figure 9 shows a small design where the two inputs a and b are added to produce the sum y . There are registers on all inputs and outputs. The di_valid signal is high when a and b are valid and the do_valid signal is high when the output y is valid.

Suggest a change to the use of di_valid and do_valid in order to reduce the energy consumption of the design. Explain why your suggestion is a good idea and why the energy consumption is reduced. Include a new version of Figure 9 as part of your answer.

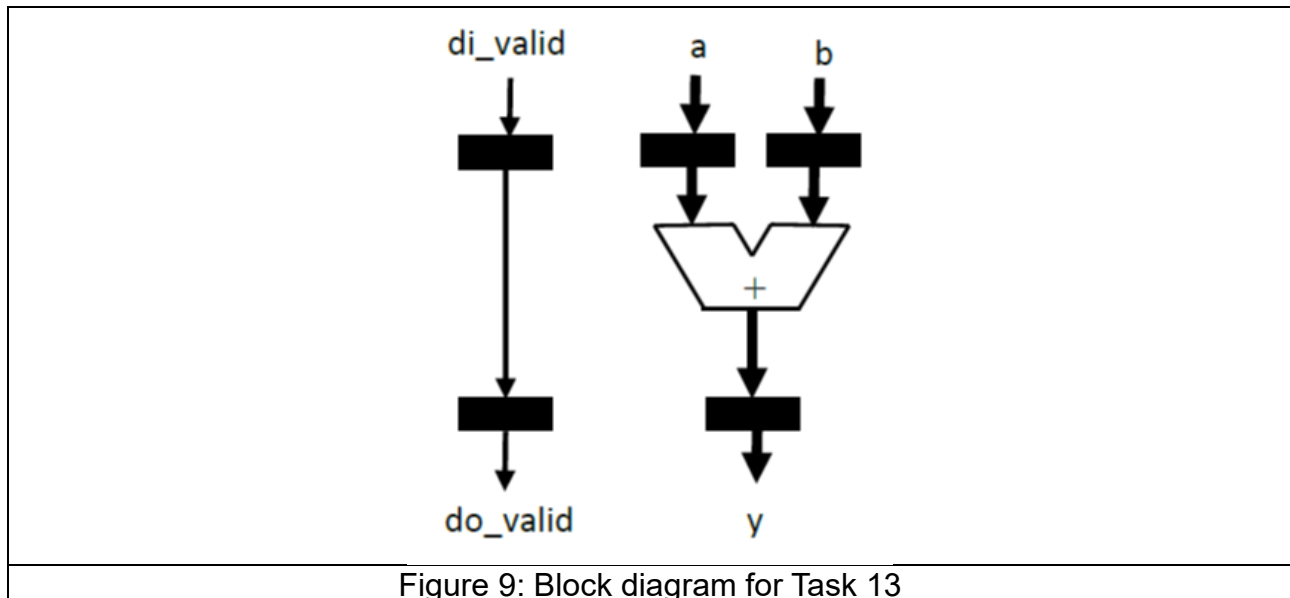


Figure 9: Block diagram for Task 13

Task 14: Coding of state machines**a) (Maximum 7 points)**

Given the code for a state machine in Figure 10. Point out at least two errors in the code. With errors we here mean that the code does not follow standard practice for state machine coding, which will result in errors during compilation and/or that the synthesis not necessarily will be as expected. Explain with a few words why these are errors.

Do NOT consider that the state machine actually does not perform a sensible function.

b) (Maximum 6 points)

Change the entity definition of the state machine in Figure 10 so that you through the use of a generic constant easily can instantiate state machines with any word width for the signals in_sig and out_sig . You are NOT supposed to consider how the architecture is to be implemented.


```
entity Task14_24_FSM is
port(in_sig, clock, reset: in bit;
      out_sig: out bit);
end entity Task14_24_FSM;

architecture RTL of Task14_24_FSM is
type state is (idle, active);
signal curr_state, next_state : state;
begin

CombProc: process (in_sig, curr_state)
begin
    case (curr_state) is
        when idle =>
            out_sig <= '0';
            if in_sig = '0' then
                next_state <= idle;
            else
                next_state <= active;
            end if;
        when active =>
            if in_sig = '1' then
                out_sig <= '1';
                next_state <= active;
            else
                next_state <= idle;
            end if;
        end case;
    end process CombProc;

SynchProc: process (clock)
begin
    if reset = '1' then
        curr_state <= idle;
    elsif clock'event and clock='1' then
        curr_state <= next_state;
    end if;
end process SynchProc;

end architecture RTL;
```

Figure 10: VHDL code for state machine in Task 14.

Task 15: RSA design**a) (Maximum 12 points)**

In the RSA term project, you had to implement an efficient modular multiplication. For example, your implementation would take as inputs two vectors to be multiplied, A and B, as well as the modulo vector N itself, and produce an output vector R. Explain how you did this based on the requirements of the RSA system, e.g., size of key and message, and possibly your own precalculated values.

Your explanations should be at a level of detail that would make it possible for one of the other students in the course to implement the module. Include block diagrams as you see fit. Note that it is only the modular multiplication you should explain, not the complete modular exponentiation.

b) (Maximum 12 points)

Define an entity for your modular multiplication unit that is able to handle incoming and outgoing data according to the ready valid handshake protocol used in the term project. Write a VHDL testbench that instantiates the entity, applies minimum two sets of input data and checks that the result is correct. Each data sets shall contain one A and one B value. The key should be the same for all sets.

Note that you are NOT supposed to write VHDL code for the modular multiplication itself, only the testbench. It is possible to do this even if you have not answered Task 15 a).

Task 16: VHDL Design

Together with the sketch in Figure 11, you have received the following module specification from your boss:

- Data arrives on a four-bit wide input bus, inbus of type `std_logic_vector`.
- The input bus is driven by a register which is clocked by the positive flank of the signal `clk`, but this is not part of your module (see Figure 11). You also have this clock signal available, and it is safe to assume that setup and hold times are complied with if you use the positive clock flank in your design.
- The module shall continuously compare the current value of each bit on the input bus with the value of the same bit during the previous clock period.
- If the two bits are equal, the corresponding bit of a four-bit `std_logic_vector` output bus, outbus, is to be set high. Otherwise it should be set low.
- The circuit shall, for example, compare the current `inbus(2)` and `inbus(2)` from the previous clock period and set `outbus(2)` high if they are equal and low if they are different. Similarly for the other bits of inbus and outbus.
- If three or more bits are different, a separate `std_logic` output signal, `diff`, shall be set high.

- The result on outbus and diff will be stored in registers clocked by the same positive clock flank, but this is not part of your module (see Figure 11).

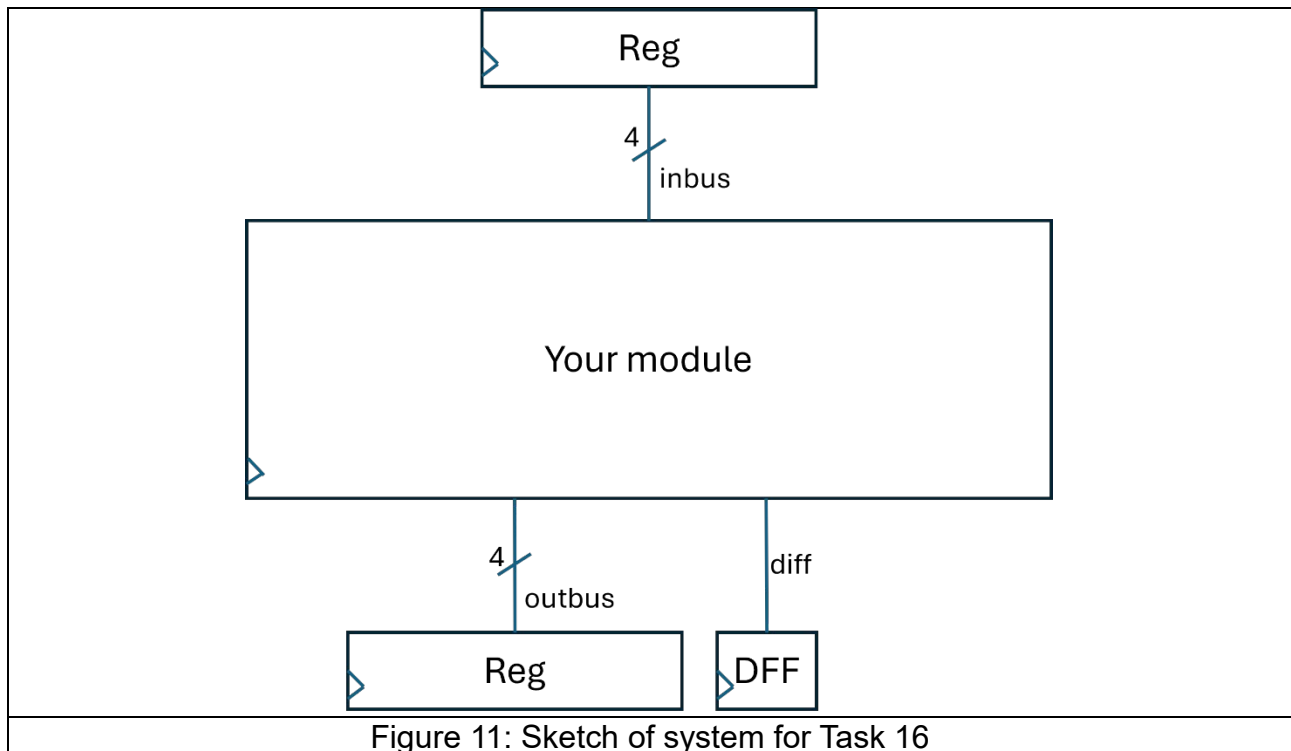


Figure 11: Sketch of system for Task 16

a) (Maximum 15 points)

Draw a microarchitecture and write VHDL-code that satisfies the specification. Include the registers and D-flipflop outside your module in the microarchitecture drawing but not in your VHDL code.

Include any assumptions you have made during the design in your answer.

b) (Maximum 7 points)

Your boss wants to know what the performance of your implementation is. Make assumptions about the different basic components in your design, e.g., related to delay and timing of registers and combinatorial logic. Then calculate the clock frequency based on your assumptions and reflect on such topics as latency and throughput.

APPENDIX A

Candidate number: _____

Write your answers to the multiple-choice tasks (1 through 12) in the table below. Detach this sheet with the table and hand it in as part of your submission.

In the example below Task 1 has been answered with alternative A. Task 2 has first been answered with alternative A and has then been corrected to Alternative C by first completely covering the X for the wrong alternative and then adding your X for the corrected alternative.

Task 1	A	X	B	C
Task 2	A		B	C X

Remember to write your candidate number on this sheet.

Task 1	A		B		C	
Task 2	A		B		C	
Task 3	A		B		C	
Task 4	A		B		C	
Task 5	A		B		C	
Task 6	A		B		C	
Task 7	A	B	C		D	
Task 8	A	B	C		D	
Task 9	A	B	C		D	
Task 10	A	B	C		D	
Task 11	A	B	C		D	
Task 12	A	B	C		D	